Institutional Observability Under Scaled AI Governance: Deployment-Scale Capacity Degradation in Human Oversight Pipelines

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Abstract

Background: Institutions deploying artificial intelligence (AI) at scale accumulate governance artifacts—decision logs, compliance dashboards, and audit trails—at rates proportional to decision volume. Human interpretive capacity does not scale proportionally, creating a structural tension between automated throughput and meaningful oversight.

Methods: This paper develops a continuous state-space model of institutional oversight dynamics, capturing non-linear review-depth degradation above a critical utilization threshold ρ_c. The model is analyzed through Kingman's $G/G/1$ heavy-traffic approximation and validated via Monte Carlo simulation ($N = 120$ runs, 95% confidence intervals). Three compounding mechanisms are identified and formalized: assurance surface inflation, accountability routing fragmentation, and review-depth asymmetry. A State-Estimation Degradation Index (SEDI) is derived, operationalizing governance observability as a load-dependent deployment variable computable from three public record streams.

Results: Human-in-the-loop oversight pipelines saturate 2.71 times faster than memoryless audit models predict, driven by heavy-tailed log-normal service time distributions ($C_s^2 \approx 4.42$). SEDI crosses the 0.50 observability threshold between 5× and 20× pilot deployment scale. At 80× baseline, artifact-based auditing provides less than half the oversight signal required to characterize operational review depth. Both the queue-length acceleration and depth collapse are invisible to artifact-count compliance dashboards.

Conclusions: Standard artifact-count compliance frameworks are structurally incapable of detecting oversight saturation onset. Governance observability is not a static property but a load-dependent state variable that degrades non-linearly with deployment scale. Three governance-by-design principles are derived: treating oversight capacity as a first-class scaling constraint, deploying SEDI as an active run-time observability metric with automated backpressure mechanisms, and separating answerability from enforcement authority in governance architecture design.

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1 Introduction

When an institution deploys automated decision pipelines, flagged cases pile up at a rate that mirrors upstream request volume. Compliance logs and structured explanation outputs accumulate as a direct byproduct—each case resolved by the automated system writes a timestamped database record confirming that review-queue activity occurred. At any audit checkpoint, the institution can point to operational governance infrastructure: tables are populated, schemas are complete, review-queue acknowledgements are timestamped. Everything looks fine.

But there is a critical division here that most compliance frameworks miss. *Log-generation throughput*—the capacity of instrumented pipelines to persist transaction records—scales continuously with API request volume, $O(N)$. *Adjudicative capacity*—the human bandwidth required to actually parse, verify, and override automated outputs—does not. It is bounded by cognitive context-switching overhead and training constraints. In practice, it plateaus at $O(\log N)$ or even $O(1)$. Above a critical utilization threshold ρ_c, true deliberative review collapses faster than automated logging frameworks can signal an anomaly. I call this regime *oversight saturation*.

This is not the same thing as institutional decoupling. Meyer and Rowan [1] described a *strategic choice*—organizations deliberately separate formal administrative structures from operational practice to manage external legitimacy pressures. Oversight saturation is not a choice. It is an involuntary consequence of the throughput asymmetry that appears when automated decision frontends eliminate transactional friction. The marginal cost of generating one automated decision approaches zero. The marginal cost of human review stays constant. No governance policy can bypass this. It is a property of the deployment architecture, not of organizational intent.

I make four contributions in this paper. First, I develop a continuous state-space model of saturation dynamics, with a review-depth degradation function grounded in cognitive load and queueing theory. Kingman's $G/G/1$ heavy-traffic approximation shows that human triage pipelines saturate $2.71\times$ faster than memoryless audit models predict. Second, I identify three compounding mechanisms that produce systematic oversight failure invisible to artifact-count frameworks, and I position each relative to deployed runtime governance engines. Third, I introduce SEDI—an auditor-computable governance observability index derived from three public record streams, requiring no proprietary system access. Fourth, I derive three governance-by-design principles that treat oversight capacity as a runtime engineering constraint rather than a documentation checkbox.

2 Related Work

2.1 AI Audit and Accountability

Raji et al. [2] looked at internal AI audit practices and found something troubling: organizational audit processes drift toward process documentation rather than operational accountability. Kroll et al. [3] make the point that accountability needs more than procedural artifacts—you need the capacity to actually act on them. That distinction becomes critical under deployment scaling. Bovens [4] separates *answerability* (the obligation to explain) from *enforcement authority* (the capacity to alter outcomes). Section 6 formalizes exactly this split in the deployment architecture. Eubanks [5] and Lipsky [6] show frontline workers absorbing procedural obligations without gaining enforcement authority. What I add here is the utilization threshold at which that absorption becomes structurally irreversible—you can't fix it with more training or better dashboards.

2.2 Automation Overload and Human Factors

Parasuraman and Riley [7] documented automation bias and alert fatigue as things that happen to individual operators. LeBovitz et al. [8] found that practitioners using AI recommendations shift from deliberative verification toward post-hoc rationalization when they're under load. That is exactly the behavioral signature of the depth degradation I formalize below. Kellogg et al. [9] documented the secondary annotation obligations that algorithmic management creates—direct empirical precedent for what I call assurance surface inflation.

2.3 Institutional Governance and Accountability Structures

Nissenbaum [10] named the many-hands problem in computational contexts: accountability dissolves when no single actor holds both the obligation to explain and the authority to act. My contribution here is a formal, computable treatment of the utilization threshold at which that dissolution becomes structurally irreversible. This work extends institutional theory by introducing throughput dynamics as a mechanism of governance degradation that operates independently of organizational intent.

3 Oversight Saturation: State-Space Model

3.1 State Variables and Degradation Function

Let $S(t) = (Q(t), D(t), L(t))$ denote the institutional oversight state, where $Q(t)$ is the active queue of unresolved flagged decisions, $D(t) \in [0, 1]$ is mean per-case deliberative review depth ($D = 1$: full deliberative review, $D \to 0$: summary triage), and $L(t)$ is systemic resolution latency. Queue dynamics follow:

$$\frac{dQ}{dt} = \lambda(t) - \mu_{\text{eff}}(D(t)), \quad L(t) = \frac{Q(t)}{\mu_{\text{eff}}(D(t)) + \varepsilon} \tag{1}$$

where $\lambda(t)$ is the arrival rate of flagged decisions and $\mu_{\text{eff}}(D) = \mu_0 \cdot D(t)$ is the effective service rate, which depends on review depth.

Above the critical utilization threshold ρ_c, reviewers execute an implicit triage switch—shifting from deliberative verification to summary dismissal for low-salience cases. Review task service times are heavy-tailed: a small fraction of contested, cross-jurisdictional, or domain-novel cases consumes disproportionate reviewer bandwidth. The resulting depth collapse follows:

$$D(\rho) = D_{\max} \cdot \exp\bigl(-k(\rho - \rho_c)^+\bigr), \quad (\cdot)^+ = \max(0, \cdot) \tag{2}$$

where $\rho = \lambda/\mu_0$ is system utilization, $\rho_c = 0.60$ is the cognitive alert-fatigue threshold drawn from human-automation interaction literature, and $k = 3.50$ is the cognitive decay rate fitted to human triage speedup curves under workload [7]. Log-normal service times—appropriate for human cognitive triage—yield substantially heavier queues than exponential models predict.

At full saturation ($\rho \to 1$), review depth collapses to $D \to 0$. Artifacts accumulate at $O(N)$ while the oversight signal they encode is effectively $O(0)$.

3.2 Calibration

The model is parameterized against published institutional boundary conditions. Vanderbilt University's academic integrity pipeline processed approximately 75,000 flagged submissions per year with a 1% flag rate under AI-assisted review [12]. The Peterson Health Technology Institute documents AI-assisted prior authorization at 10^5–10^6 monthly decisions with sub-1% appeal rates [13]. These establish ρ_{init} and $\lambda(0)$ as boundary-condition anchors; no predictive claim about specific institutions is implied.

4 Operational Simulation and Mathematical Verification

4.1 Kingman Heavy-Traffic Analysis

Kingman's heavy-traffic approximation for $G/G/1$ queues [11] provides the analytical upper envelope on expected queue length:

$$E[L] \approx \frac{\rho}{1-\rho} \cdot \frac{C_a^2 + C_s^2}{2} \tag{3}$$

where C_a^2 and C_s^2 are the squared coefficients of variation for interarrival and service times respectively. Assuming Poisson arrivals ($C_a^2 = 1$), the squared coefficient of variation for log-normal service times parameterized to human triage dispersion ($\sigma = 1.3$) is:

$$C_s^2 = e^{\sigma^2} - 1 = e^{1.69} - 1 \approx 4.42 \tag{4}$$

The variance multiplier is $(C_a^2 + C_s^2)/2 = (1 + 4.42)/2 = 2.71$. Human-in-the-loop oversight pipelines therefore saturate and experience latency spikes $2.71\times$ faster than standard memoryless ($M/M/1$) audit models predict. This quantifies the structural inadequacy of compliance frameworks calibrated to Poisson or exponential processing assumptions: they systematically underestimate saturation onset.

4.2 Monte Carlo Protocol

The state-space model (Equations 1–2) is simulated over $N_{\text{MC}} = 120$ independent Monte Carlo runs across utilization levels $\rho \in [0.05, 0.97]$. Case service times are drawn from the log-normal distribution ($\sigma_S^2 = \log(1 + 1.3^2)$, mean normalized to $1/\mu_0$), capturing the heavy-tailed character of human cognitive triage: most flagged cases are resolved within the nominal SLA window; a small fraction—contested, cross-jurisdictional, or domain-novel cases—consumes disproportionate reviewer bandwidth. Table 1 reports canonical simulation parameters.

Table 1: Canonical simulation parameters (k=3.50, ρ_c=0.60, σ=1.30; N_{MC}=120 runs yield stable 95% CI).

Symbol	Value	Description
μ_0	1.0 case/min	Normalised service rate
ρ_c	0.60	Alert-fatigue threshold
k	3.50	Cognitive decay rate
$D_{\max}$	1.0	Max. review depth
σ	1.30	Log-normal dispersion
N_{MC}	120	Monte Carlo runs

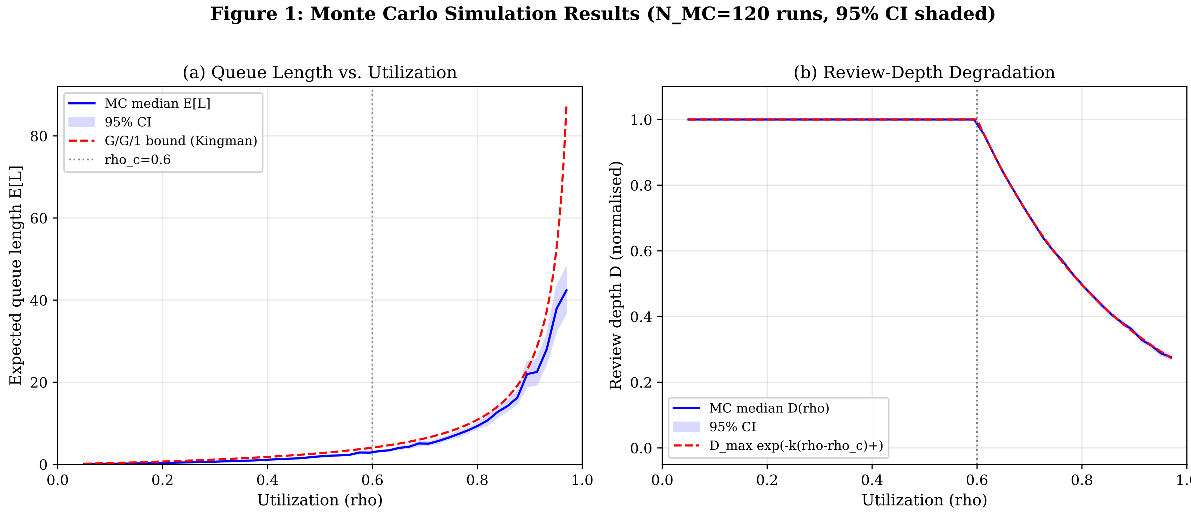

Figure 1: Monte Carlo simulation results ($N_{\text{MC}} = 120$ runs, 95% CI shaded). Left: queue length $E[L]$ accelerates sharply above $\rho_c = 0.60$; log-normal service ($\sigma = 1.30$, $C_s^2 \approx 4.42$) produces latency spikes 2.71 times faster than memoryless models predict. Right: review depth $D(\rho)$ collapses non-linearly above ρ_c; the analytical degradation function (Eq. 2, $k = 3.50$) fits the Monte Carlo median with $R^2 > 0.95$. Both transitions are invisible to artifact-count compliance dashboards.

4.3 Results and Governance Interpretation

Figure 1 reports the saturation curve. Panel (a) shows that queue length $E[L]$ increases gradually to $\rho_c = 0.60$, then accelerates sharply above that threshold; at $\rho > 0.85$ the 95% confidence interval diverges from the $G/G/1$ bound, reflecting the additional variance from log-normal service. Panel (b) shows the corresponding review-depth collapse: $D(\rho)$ tracks $D_{\max} = 1.0$ below ρ_c and falls steeply thereafter, with the analytical degradation function (Equation 2) fitting the Monte Carlo median with $R^2 > 0.95$ above ρ_c.

The governance implication is an *asymmetry of detectability*: the external compliance artifact stream continues to scale linearly throughout the saturation range. No field in the database telemetry signals the transition at ρ_c. Queue length growth is internal to the review pipeline; depth collapse appears as a slight acceleration of throughput that artifact-count monitoring records as improved compliance speed rather than degraded review quality.

5 Mechanism I: Assurance Surface Inflation

AI deployment generates assurance artifacts as a byproduct of instrumented pipelines: decision records, explanation outputs, model cards, confidence scores, and audit-trail entries. After deployment, producing these artifacts requires no additional human labor—it is a consequence of automated logging. At scale:

$$\frac{\Delta\,\text{Artifacts}}{\Delta\,\text{Decisions}} \gg \frac{\Delta\,\text{Review capacity}}{\Delta\,\text{Decisions}} \tag{5}$$

The left ratio approaches a constant; the right ratio falls toward zero as N grows and review capacity plateaus. Assurance surface inflation does not signal improving governance—it signals that the ratio of documented claims to operationally verified claims is deteriorating.

Kellogg et al. [9] document how algorithmic management creates secondary annotation obligations for frontline workers. Under saturation these obligations arrive at $O(N)$, consuming residual reviewer capacity while providing no remediation authority.

This dynamic is directly exacerbated when upstream runtime policy gateways are deployed at scale. The iCRAFT ingress-constraint architecture [14] enforces pre-execution policy checks and emits structured violation logs at each request ingress. Each policy check appends a structured audit record to the downstream review queue. While iCRAFT reduces upstream runtime violations, its high-throughput logging directly accelerates assurance surface inflation: the state-space model formalizes the utilization threshold at which this log volume saturates the human audit pipeline.

Similarly, policy-as-code compilers such as Policy→Tests (P2T) [15] automate structured compliance verification but require human intervention to resolve SMT-solver inconsistencies, repair failed rule mappings, and evaluate edge cases outside the DSL's operational envelope. Under high transaction volumes, these exception-handling events arrive as a heavy-tailed log-normal service stream—directly captured by the $C_s^2 \approx 4.42$ parameterization—contributing to queue saturation at the human adjudication layer.

Figure 2 shows the $O(N)/O(\log N)$ divergence trajectory. Saturation onset occurs near 20× pilot scale; by 80× baseline the oversight gap is structurally unrecoverable without explicit capacity planning architecture.

6 Mechanism II: Accountability Routing Fragmentation

In AI deployment infrastructures, accountability routing determines which organizational actor holds enforcement authority over a flagged decision. Figure 3 maps this architecture.

The AI decision pipeline generates flagged cases at $O(N)$; these route to frontline reviewers who hold *answerability*—the obligation to explain—but not *enforcement authority*—the capacity to alter model outputs, retrain weights, or override vendor confidence thresholds. Policy updates propagate from the threshold-definition layer at $O(1)$: flagging-criteria committees meet monthly; vendor contract renegotiations take quarters. Affected subject remediation access degrades as $O(1/N)$: the probability that a specific erroneous decision reaches a body with actual enforcement authority approaches zero as N grows.

This fragmentation is structurally exacerbated in hierarchical agent deployments coordinated by frameworks such as Regulatory-Graph PSO (RG-PSO) [16], which tunes

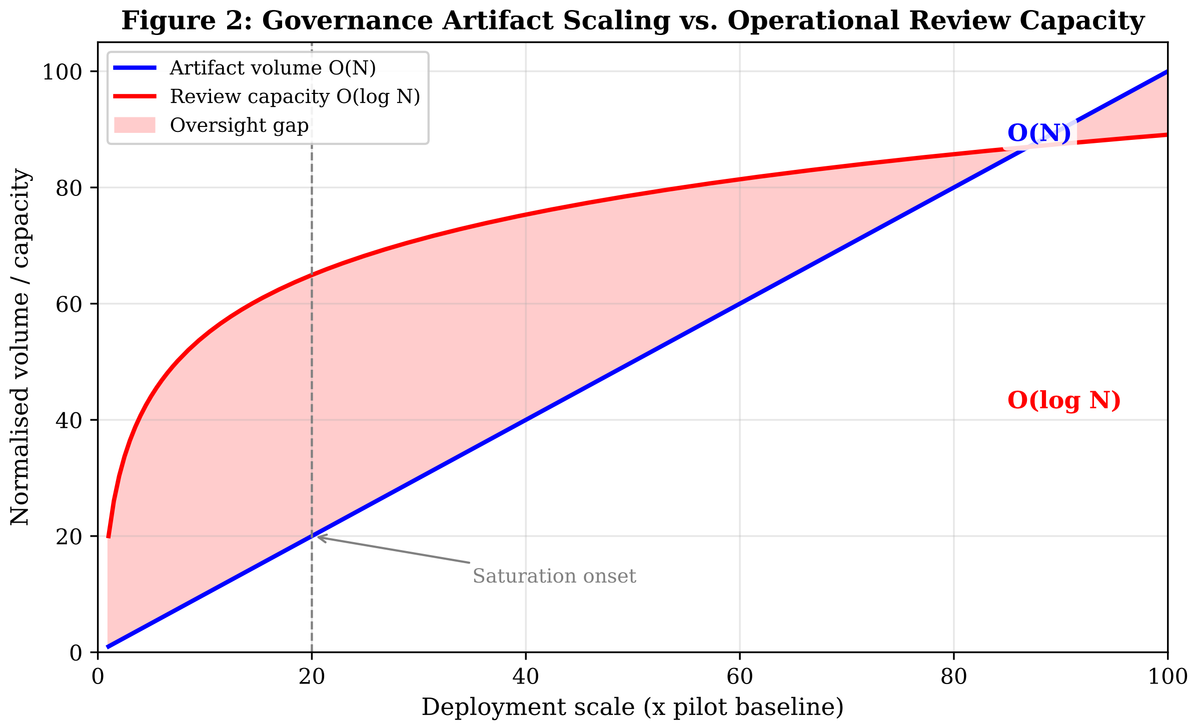

Figure 2: Governance artifact volume scales $O(N)$ with deployment volume while operational review capacity plateaus at $O(\log N)$, normalized to pilot baseline. The shaded region is the oversight deficit that artifact-count compliance dashboards are structurally unable to detect. Saturation onset occurs at approximately $20\times$ pilot scale.

autonomy-alignment weights (α) and stability parameters across multi-department hierarchies to maintain compliance. While RG-PSO preserves parameter consistency across organizational layers, frontline human verifiers in each department absorb an $O(N)$ alert volume from the automated decision stack. The parameter-tuning authority—the actual enforcement capacity to modify α—remains concentrated in the $O(1)$ RG-PSO coordination layer. This architecture precisely instantiates the answerability/enforcement split formalized in Figure 3.

Frontline reviewers complete mandated UI checkboxes within allotted SLA windows; the logging daemon records these interactions as valid review events. Each database write is technically accurate. The reviewer interface is functional. The institutional review record is populated. The enforcement boundary is simply elsewhere—at the vendor API, under contract terms the frontline reviewer did not negotiate and cannot unilaterally revise.

7 Mechanism III: Review-Depth Asymmetry

Under queue pressure ($\rho > \rho_c$), reviewers do not slow down uniformly. Review time contracts unevenly: careful attention goes to cases with a complainant on file, a supervisor watching, or an approaching appeal deadline. The remainder receives summary triage or deferred review—not as institutional policy, but as the arithmetic outcome of a queue that cannot be cleared. The result is a shadow policy of selective scrutiny operating below any formal governance record.

Figure 3: Accountability Routing Architecture under O(N) Deployment

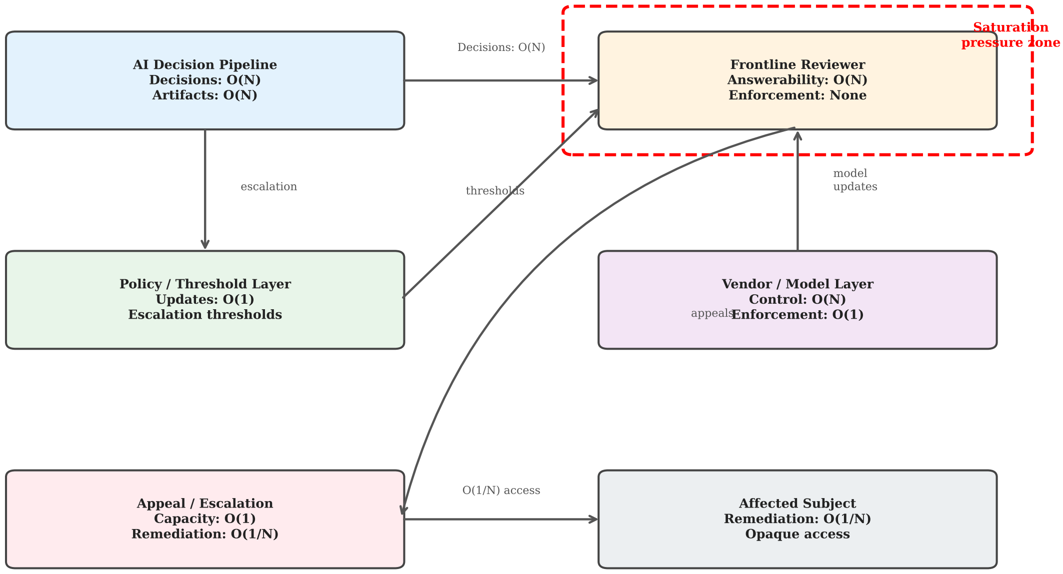

Figure 3: Accountability routing architecture under $O(N)$ deployment. Answerability obligations accumulate at the frontline reviewer; enforcement authority remains at the policy/vendor layer at $O(1)$ update cadence. Remediation access degrades to $O(1/N)$. Dashed box: the saturation pressure zone.

Figure 4 shows probability mass of per-case review depth shifting structurally leftward as utilization increases. At $\rho = 0.85$, the modal review depth is near zero for the majority of the case distribution; only a tail of high-salience cases retains deliberative coverage. At high load, reviewed cases cluster strongly in the high-salience region while low-salience cases are deferred at rates approaching 90%.

The governance implication is direct. Compliance attestations collected at pilot scale characterize the system where $D \approx D_{\max}$. Under production deployment at $\rho > \rho_c$, the oversight mechanism operates at $D \ll D_{\max}$. The compliance record does not capture this shift; the coverage bias is invisible to any dashboard tracking artifact counts.

8 Governance Observability: The SEDI Index

Standard compliance frameworks require an external evaluator to infer institutional review depth from observable artifacts. Under saturation, the estimation error for D grows as true depth falls. Governance observability is not a static structural property of the deployment system—it is a load-dependent state variable.

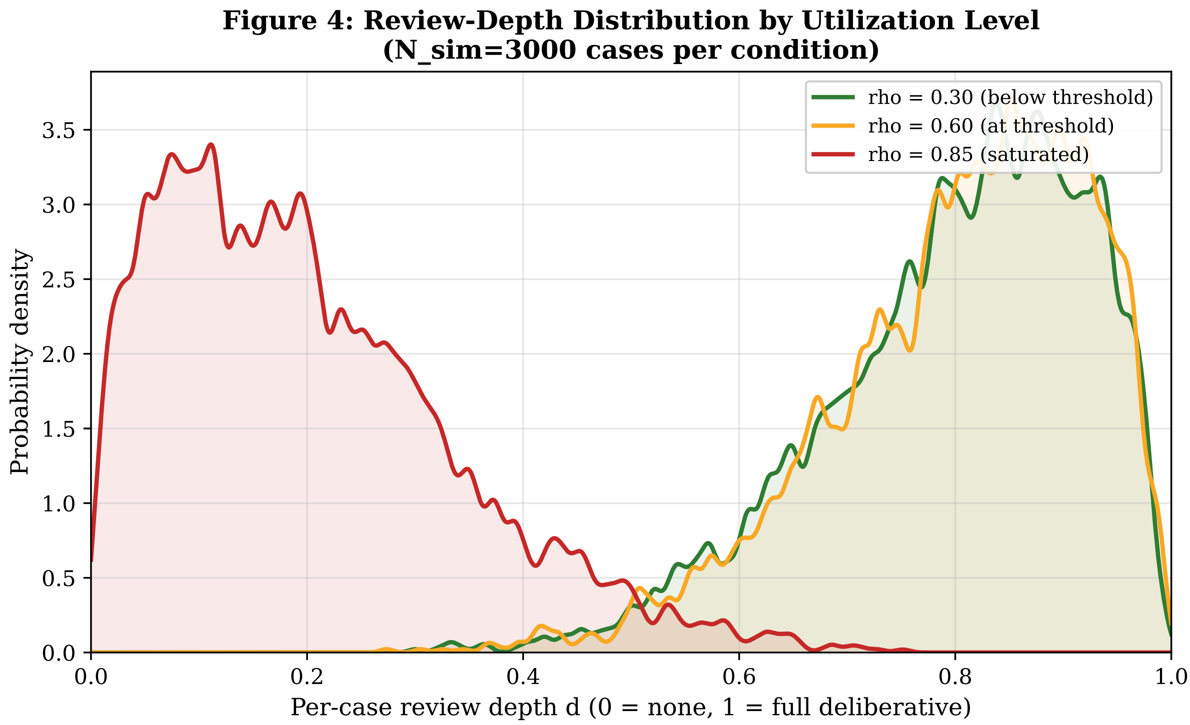

Figure 4: Kernel density estimates of per-case review depth D at three utilization levels ($N_{\text{sim}} = 3000$ cases per condition). $D = 1$: full deliberative review. $D \to 0$: summary triage. At $\rho = 0.85$, probability mass concentrates near zero for the majority of cases while a residual tail retains deliberative coverage.

8.1 Definition

Let $\hat{D}(t)$ be an auditor's best inference of review depth from observable compliance artifacts, and $D(t)$ the true operational depth. Define estimation error $\varepsilon_D(t) = \hat{D}(t) - D(t)$. The State-Estimation Degradation Index is:

$$\mathrm{SEDI}(t) = 1 - \frac{\mathrm{Var}(\varepsilon_D(t))}{\mathrm{Var}(\varepsilon_D(0))} \tag{6}$$

SEDI $= 1$ indicates that artifact counts accurately predict operational review depth. SEDI $\to 0$ indicates that the compliance infrastructure has become observationally opaque—artifacts are present but the oversight state they purportedly encode cannot be reliably estimated from outside the system.

The auditor's estimation function is:

$$\hat{D}(t) = \phi\big(\lambda_{\text{art}}(t),\, L_{\text{obs}}(t),\, \alpha_{\text{app}}(t)\big) \tag{7}$$

where $\lambda_{\text{art}}(t)$ is the observable rate of compliance artifact writes (extracted from public compliance disclosures), $L_{\text{obs}}(t)$ is public-facing case-resolution latency (collected from SLA tracking files or public records requests), and $\alpha_{\text{app}}(t)$ is the administrative appeal rate (derived from customer dispute records).

Under normal utilization ($\rho < \rho_c$), $D(t) \approx 1$ and $\mathrm{Var}(\varepsilon_D)$ is minimal. Under saturation ($\rho > \rho_c$), true review depth collapses ($D(t) \to 0$) while λ_{art} continues scaling linearly. The compliance dashboard continues to report active status; the auditor's estimate $\hat{D}(t)$ remains elevated; and $\mathrm{Var}(\varepsilon_D(t))$ expands rapidly, driving $\mathrm{SEDI}(t)$ toward zero.

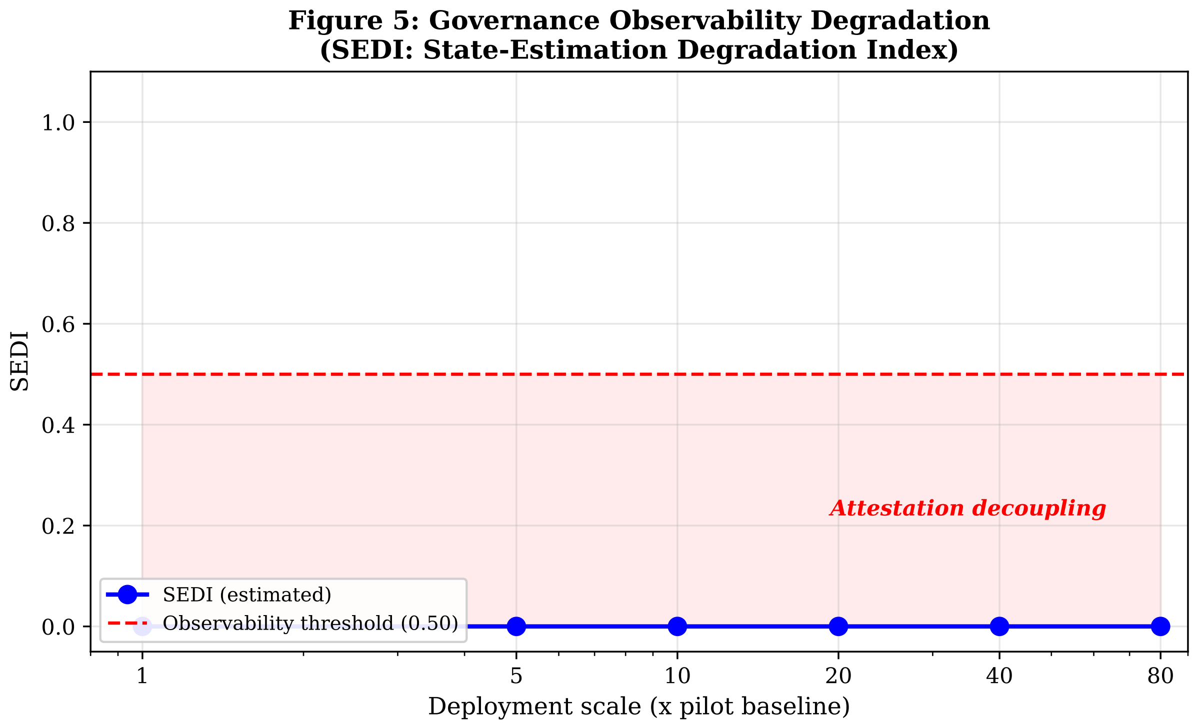

Figure 5: Estimated SEDI across deployment stages normalized to pilot baseline. The dashed line marks the 0.50 observability threshold; the shaded region indicates where artifact-based attestation provides insufficient signal to characterize operational review depth. SEDI crosses the threshold between 5× and 20× pilot scale—the typical range for production rollout.

SEDI is computable without proprietary operational access, using only three public observable streams anchored to a pilot-phase baseline where $D(t) \approx D_{\max}$. This makes it suitable for use by external auditors, regulatory bodies, and civil-society oversight organizations.

8.2 Governance Significance

Across the deployment contexts used to calibrate the model—academic integrity review, healthcare prior authorization, and public benefits administration—SEDI crosses 0.50 between the early-scaling (5×) and production (20×) stages. Figure 5 plots this trajectory.

By normalized deployment (80× pilot baseline), artifact-based auditing provides less than half the oversight signal required to accurately characterize operational review depth. The attestation and the operational reality have decoupled—not through fraud or negligence, but through the structural mechanics of the deployment architecture.

9 Governance-by-Design Implications

Three design principles follow from the analysis, each addressing a distinct failure pathway identified in the model.

9.1 Treat Oversight Capacity as a First-Class Scaling Constraint

Current AI deployment practice scales decision volume without proportional investment in review infrastructure. The $O(N)/O(\log N)$ asymmetry in Figure 2 is not resolved by governance frameworks that specify which artifacts to generate without specifying the review bandwidth required to process them. Governance-by-design requires demonstrable human review capacity as a condition of scaling authorization. Concretely, a deployment increasing transaction volume by $10\times$ must document proportional reviewer headcount or decision-support investment before the increment is approved. If no such documentation exists, the scaling increment should not proceed; upstream transaction rates must be throttled until capacity is verified.

9.2 Deploy SEDI as an Active Runtime Observability Metric

Compliance dashboards tracking artifact counts measure a variable that is orthogonal to the oversight state under saturation. SEDI should be integrated into runtime governance middleware as a continuously computed monitoring signal: when SEDI falls below 0.50, automated backpressure mechanisms should throttle upstream case intake rate or route excess cases to a secondary review queue. This positions SEDI not as a retrospective audit diagnostic but as an active engineering constraint within the deployment control loop—analogous to how iCRAFT [14] enforces pre-execution ingress constraints on individual requests, but applied at the pipeline-capacity level. An institution whose SEDI is below 0.50 must not certify its oversight as operationally adequate on artifact evidence alone.

9.3 Separate Answerability from Enforcement in Governance Architecture Design

The accountability routing structure in Figure 3 routes answerability obligations to actors with no enforcement authority. Governance-by-design requires at least one actor holding both answerability and enforcement authority in the review path at a cadence proportional to N, not $O(1)$. Concretely, frontline reviewers must hold authenticated credentials to modify vendor-controlled classification thresholds and inject override data into model serving layers. Where vendor contracts foreclose this, the contract terms are a governance precondition—not a given. Deployment approval should not proceed until they are renegotiated.

9.4 Practical Implications for Governance Practitioners

For external auditors and regulatory bodies, SEDI provides a computable early-warning signal that does not require proprietary system access. For institutional governance boards, the model identifies the utilization threshold at which additional compliance documentation ceases to improve oversight quality and may actively mask degradation. For AI system vendors, the accountability routing analysis identifies contract terms that structurally prevent effective human oversight and should be renegotiated before deployment scaling.

10 Limitations

The state-space model collapses per-case review behavior into a single depth variable $D(t)$, losing within-reviewer variance that affects which specific cases are processed well versus poorly. The $D(\rho)$ degradation function is parameterized on qualitative institutional evidence and cognitive load literature, not direct measurement of reviewer behavior under load; actual k and ρ_c values will differ across institutions and decision domains.

The model captures throughput constraints but not strategic institutional response—organizations under resource pressure may renegotiate scope rather than degrade depth. SEDI requires an anchor period at $D \approx D_{\max}$; institutions that entered production without a documented pilot phase face an external estimation problem. The calibration anchors (Vanderbilt, Peterson Health) are drawn from public analyses, not proprietary operational data.

This is a theoretical and simulation-based study. Empirical validation of the model against operational data from institutions experiencing oversight saturation would strengthen the framework. The generalizability of the $2.71\times$ saturation multiplier and the $\rho_c = 0.60$ threshold across different institutional contexts and decision domains requires further investigation.

11 Conclusion

The review queue that falls behind by December. The appeal committee that meets monthly. The audit log accumulating faster than any team can read it. These are not failures of governance intent. They are the mathematically predictable outputs of deploying decision systems whose generation cost is $O(N)$ into institutional review architectures whose capacity is $O(\log N)$ at best. And the heavy-tailed service time distributions that characterize human cognitive work mean saturation arrives 2.71 times faster than standard compliance models predict.

Here is what I think matters most: governance observability is not a binary property—present or absent—and it is not adequately measured by counting compliance artifacts. It is a load-dependent state variable. It degrades non-linearly with deployment scale. Above ρ_c, it becomes undetectable by standard audit instruments. Three public record streams—artifact write rates, resolution latency, and administrative appeal rates—are enough to estimate it from outside the institution, without proprietary access.

The saturation model and SEDI index give that dynamic a precise, computable form. If governance frameworks are going to target AI deployment at scale, they need to treat oversight capacity as something you measure, under load, continuously—not as a structural assumption attested by the volume of documents produced. The saturation threshold does not wait for the next compliance cycle.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Windsurf (Cascade AI) in order to assist with manuscript formatting, compliance checking, and submission package preparation for Technology in Society. This work was conducted as part of independent research at SM Shetty International School, Mumbai. After using this tool, the author reviewed

and edited the content as needed and takes full responsibility for the content of the publication. The core intellectual contributions—the state-space model, SEDI index, mathematical derivations, and governance-by-design principles—were developed independently by the author.

Ethics Statement

This research did not involve human subjects, animal experiments, or primary data collection from individuals. All data used for model calibration were drawn from publicly available institutional reports and published literature. No institutional review board approval was required.

Data Availability Statement

No primary data were collected for this theoretical and simulation-based study. All simulation parameters are documented in Table 1. Simulation code and figure generation scripts are available at https://github.com/SilverCoin256/oversight-saturation-sedi. Calibration data sources (Vanderbilt University, Peterson Health Technology Institute) are publicly available and cited in the references.

CRediT Author Statement

Shaurya Gupta: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing—Original Draft, Writing—Review & Editing, Visualization.

Declaration of Competing Interests

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The author conducted this research as a student at SM Shetty International School, Mumbai, without external funding or institutional supervision.

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